

# Rahul Debnath

## Software Engineer & Aspiring Robotics Engineer

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Department of Electrical and Computer Engineering, RUET

## Professional Summary

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Electronics and Computer Engineering student with demonstrated leadership in technical clubs, research publications, and competitive programming. Proven track record in autonomous systems development, deep learning applications, and ROS2-based robotics. Seeking opportunities to apply multi-disciplinary engineering skills in innovative technology roles.

## Education

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### Rajshahi University of Engineering & Technology (RUET)

#### Bachelor of Science in Electronics and Computer Engineering

*Expected: July 2026*

- **CGPA:** 3.60/4.00
- **Relevant Coursework:** Control Systems & Robotics, Machine Learning, Computer Vision, Micro-processor & Assembly Language, Power Electronics, Data Structure & Algorithms

## Leadership & Extracurricular Activities

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- **Vice Chair, IEEE RUET Student Branch** *2025-2026*
- **Vice President, Astronomy and Science Society RUET** *2025-2026*
- **Vice President, Robotic Society of RUET** *2025-2026*
- **Vice President, RUET CORE** *2025-2026*
- **Joint Secretary, ECE Club RUET** *2025-2026*
- **Class Representative, Department of ECE** *2025-2026*

## Honors & Awards

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- **6th Place Finalist** - Mars Autonomous Rover Rally (NASA-supported) *2025*
- **1st Place, Poster Presentation** - RUET Innovista 2025 *2025*
- **3rd Place** - RUET University Day Research Competition *2024*
- **7th Place, National Round** - 5th Kibo Robot Programming Challenge *2023*
- **Champion** - Codespark Intra-RUET Programming Contest *2022*

## Publications

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- **Deep Learning Based Smart Bin for Waste Sorting**
  - Developed YOLOv8s-based smart bin (92.7% mAP@50) for real-time classification of recyclable, non-recyclable, and compostable waste
  - Optimized for real-time deployment with minimal processing overhead
- **ROS2 Autonomous Wheelchair with LLM-Driven Control**
  - Built ROS2 platform with LiDAR SLAM and Pure Pursuit navigation

- Integrated conversational assistant (91.5% accuracy, 0.575s response time) on Raspberry Pi 4B/ESP32
- **Multi-Controller Framework for Grid-Connected EVs**
  - Developed MATLAB/Simulink framework comparing PI, PID, SMC, and MPC controllers
  - Demonstrated SMC robustness and MPC predictive performance for EV-grid integration

## Technical Projects

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- **HallFeast:** Meal management website with full-stack implementation
- **ASSR Website:** Club activity showcase and promotion platform
- **Autonomous Mobile Robot:** ROS-based navigation and control system
- **RC Quadcopter:** Custom-built drone with flight control systems
- **LFR (Line Following Robot):** Autonomous navigation using sensor arrays

## Event Organization & Volunteering

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- **Organizing Committee Volunteer:** IEEE International Conference PEEIACON 2024
- **Event Coordinator:** IEEE Day 2025, Annual General Meetings, Seminars, and Workshops
- **Volunteer:** American Corner Pop-Up Programme by US Embassy at Department of ECE
- **Lead Organizer:** Multiple Orientation and Farewell Programmes for ECE Department

## Workshops Conducted

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- **Instructor:** Internet of Things(IOT) Workshop *Dec 2025*
- **Instructor:** Microcontroller Programming Workshop *Oct 2025*
- **Instructor:** Line Following Robot (LFR) *Feb 2025*

## Industrial Training

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- **Software Training:** BJIT *(2 Weeks)*
  - Software development methodologies and industry practices
- **Automation Engineering:** Ulterior Engineering *(1 Month)*
  - Industrial automation systems and control engineering

## Technical Skills

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|------------------------------|----------------------------------------------------------------|
| <b>Programming:</b>          | Python, C++, MATLAB, JavaScript, ROS2, Java                    |
| <b>Frameworks/Libraries:</b> | YOLOv8, TensorFlow, PyTorch, OpenCV, Simulink                  |
| <b>Hardware:</b>             | Raspberry Pi, ESP32, Arduino, Microcontrollers, Sensors, LiDAR |
| <b>Tools:</b>                | Git, Docker, Linux, LaTeX, OnShape                             |

## References

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*Supervised autonomous robotics project*

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*Academic advisor and research mentor*